

PROJECT 2 – ANALYZING POINT PATTERNS

SSCI 583: SPATIAL ANALYSIS AND MODELING

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Introduction

The goal of this project is to analyze the spatial pattern of fatal police encounters in New York City using point pattern methodologies and to produce a counter-map that reframes risk in urban space. Rather than mapping crime risk, this project visualizes and evaluates the spatial risk of fatal police violence using the Fatal Encounters dataset. Following the framework of O'Sullivan and Unwin (2010), spatial point pattern methods were implemented to determine whether fatal encounters are randomly distributed or clustered across NYC.

Three major analytical components structure this study: (1) first-order intensity analysis using Kernel Density Estimation (KDE), (2) second-order spatial dependence testing using Ripley's K function, and (3) inhomogeneity adjustment to account for varying population density across the city. A sensitivity test addressing the Modifiable Areal Unit Problem (MAUP) was also conducted using a fishnet grid.

Study Area :New York City

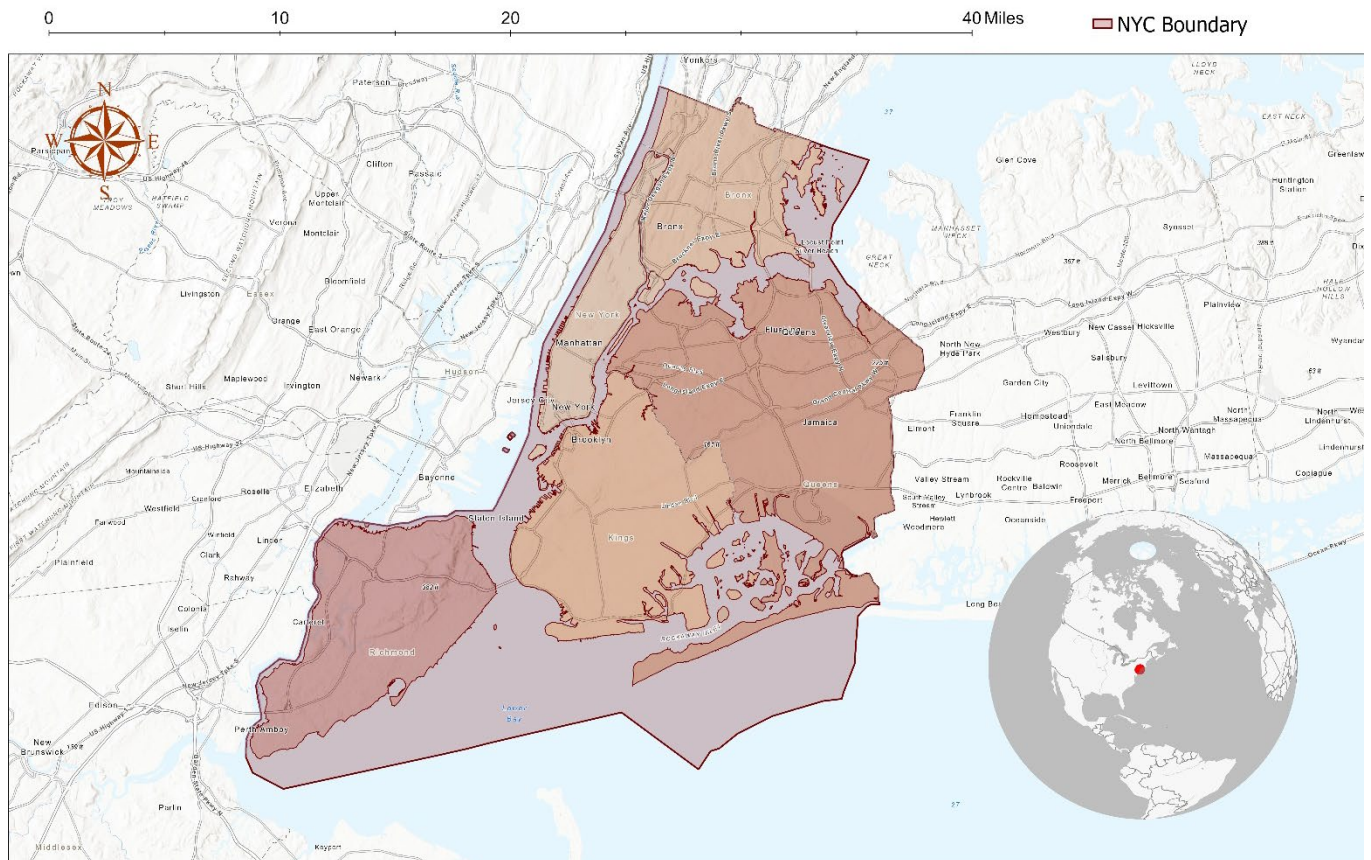


Figure 1 Study Area Map

The study area is New York City, composed of five boroughs: Manhattan, Bronx, Brooklyn, Queens, and Staten Island. All data were projected into NAD 1983 (2011) State Plane New York Long Island (US Feet) to minimize distortion in distance-based analyses.

Two foundational maps were created to contextualize the study area:

1. A borough boundary map defining the analytical extent. (figure 1)
2. A population choropleth map using 2020 census tracts, illustrating strong spatial inhomogeneity across the city, with particularly high population density in Manhattan and portions of Brooklyn and Queens.

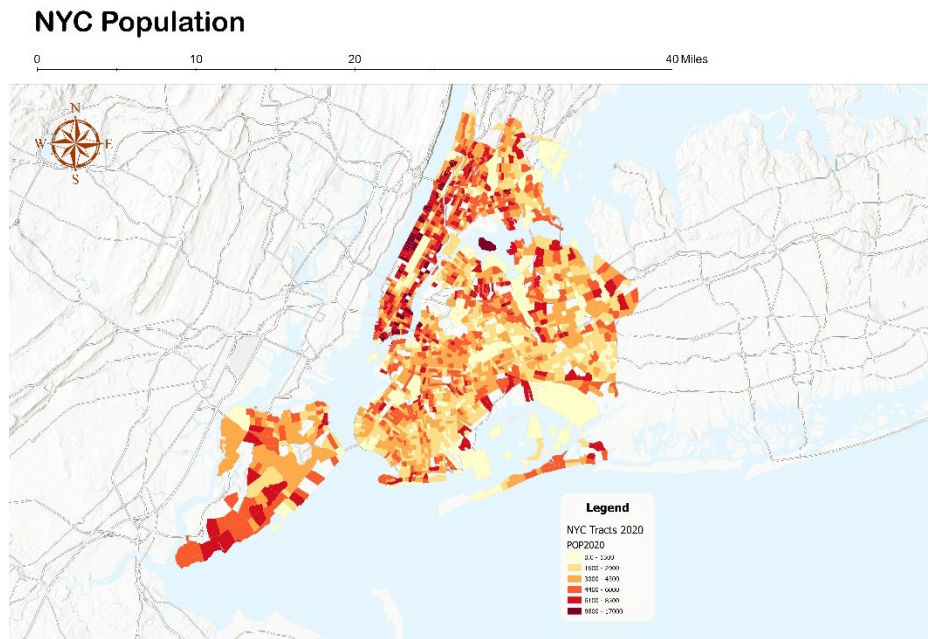


Figure 2 NYC Population (tracts)

Data Description and Table

The Fatal Encounters dataset contains geocoded incident locations and attributes derived from crowd-sourced reporting (Finch et al., 2019). While not officially comprehensive, it represents the most robust available dataset for this topic.

Table 1 Datasets

Dataset	Description	Format	Source
FatalEncounters_NYC	Fatal police encounters from 2000–present, clipped to NYC boundary	Vector point	Fatal Encounters (Finch et al., 2019)
NYC_Borough_Boundaries	Borough-level administrative polygons	Vector polygon	NYC Open Data
NYC_Boundary	Dissolved NYC boundary polygon	Vector polygon	Derived from boroughs
NYC_Census_Tracts_2020	Census tracts with 2020 population counts	Vector polygon	NYC Open Data

Methods

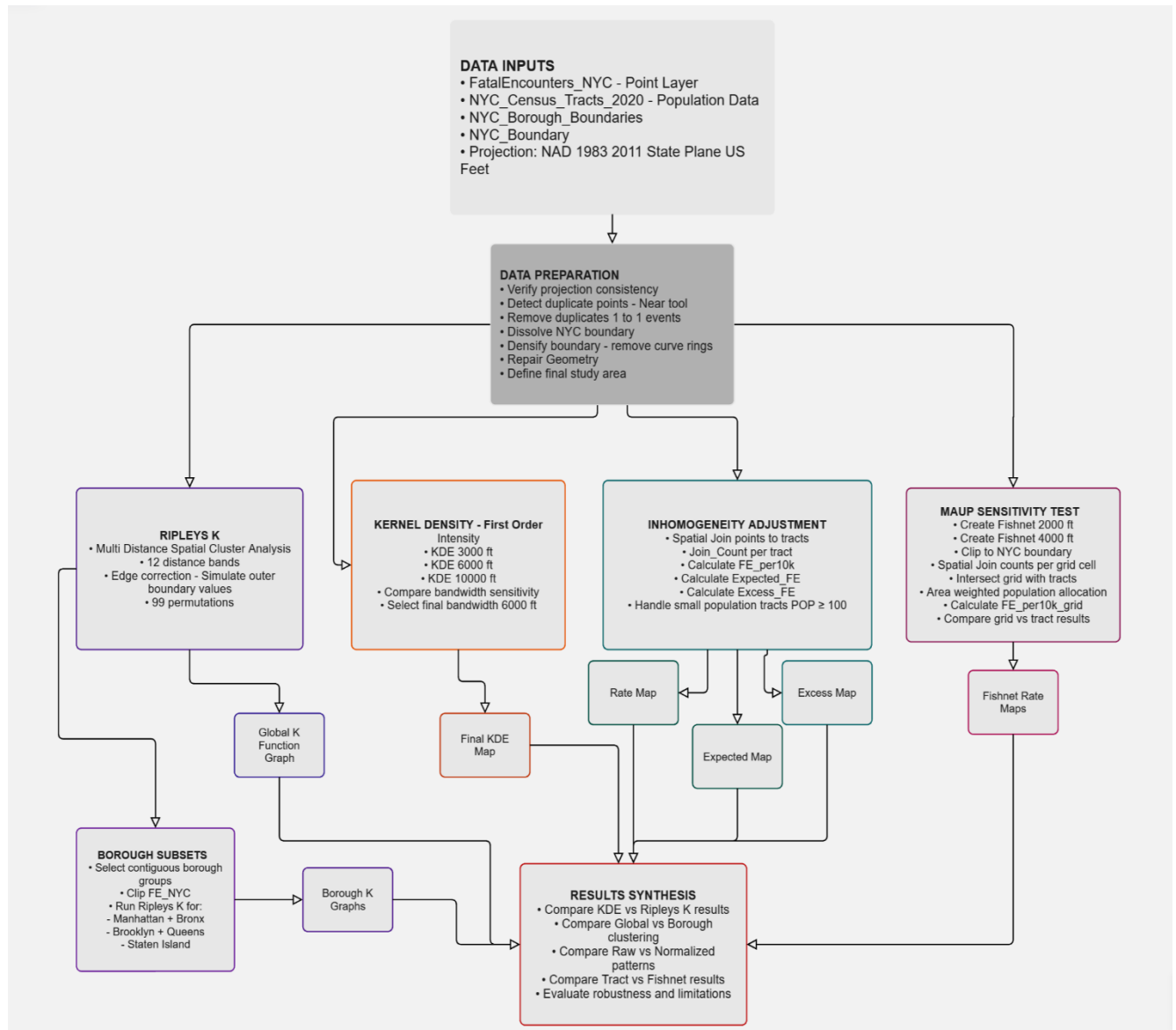


Figure 3: Workflow Diagram.

Data Preparation

Initial inspection revealed duplicate point locations. The Near tool was used to identify coincident points, and duplicates were manually reviewed to ensure a 1:1 relationship between location and event. This step was necessary because Ripley's K assumes independent events at distinct coordinates.

The study area boundary was defined using a dissolved NYC boundary. Analyses were run both globally and for borough groupings to assess the influence of water barriers.

Kernel Density Estimation (First-Order Intensity)

Kernel Density Estimation was used to visualize the spatial intensity of fatal encounters. Multiple bandwidths were tested (3,000 ft, 6,000 ft, and 10,000 ft) to assess sensitivity.

- 3,000 ft bandwidth revealed localized clusters but showed fragmented hot spots.
- 6,000 ft bandwidth produced more coherent regional clusters.
- 10,000 ft bandwidth oversmoothed and merged distinct clusters.

The 6,000 ft surface was selected as the final intensity visualization because it balanced local detail with regional interpretability.

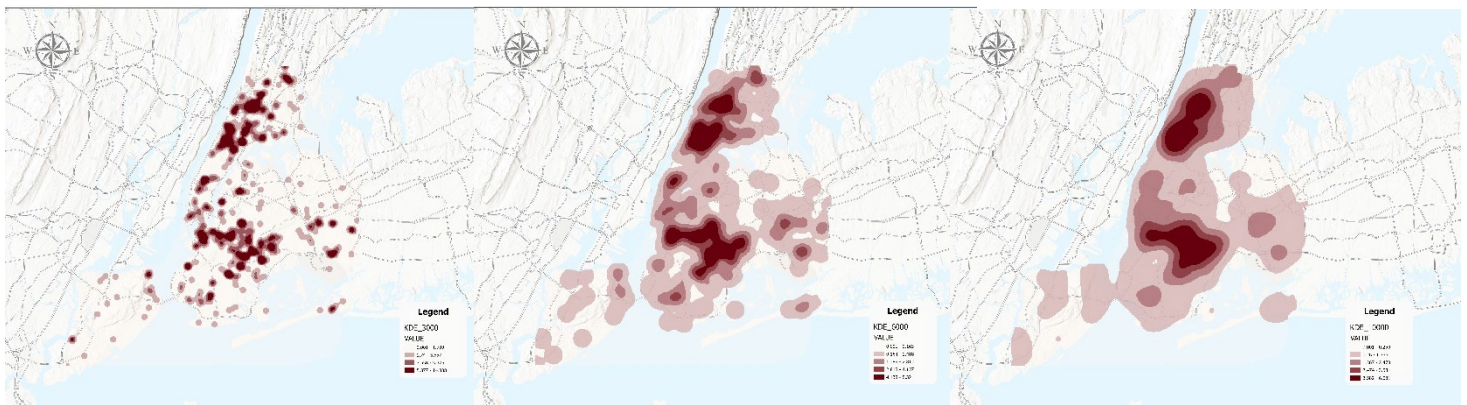


Figure 3 Kernel Density surfaces at 3,000 ft, 6,000 ft, and 10,000 ft bandwidths.

Water barriers were acknowledged as a limitation because KDE spreads intensity across space regardless of physical barriers.

Ripley's K Function (Second-Order Analysis)

Ripley's K was implemented using Multi-Distance Spatial Cluster Analysis to assess clustering across distance bands. Parameters were selected based on exploratory testing:

- Multiple distance bands
- Increment sizes selected to capture neighborhood to borough scales
- Edge correction applied (simulate outer boundary values)

- Study area defined using user-provided boundary

Global results show the Observed K curve consistently above the Expected K and above the upper confidence envelope across nearly all distance bands. This indicates statistically significant clustering.

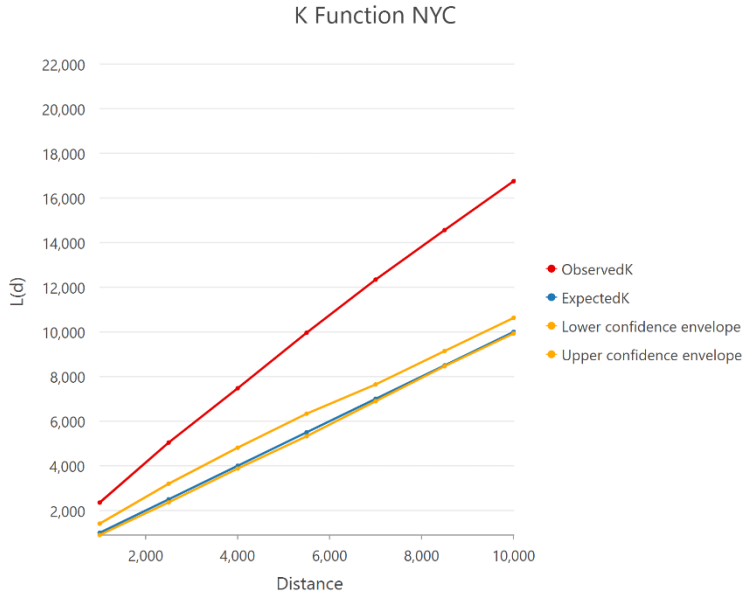


Figure 4 K NYC

Borough grouping results reveal spatial differences:

- Manhattan & Bronx show strong clustering.
- Brooklyn & Queens also show clustering, though slightly less extreme.
- Staten Island shows unstable patterns due to small sample size.

This confirms clustering is not random and varies by borough grouping, partly influenced by water separation.

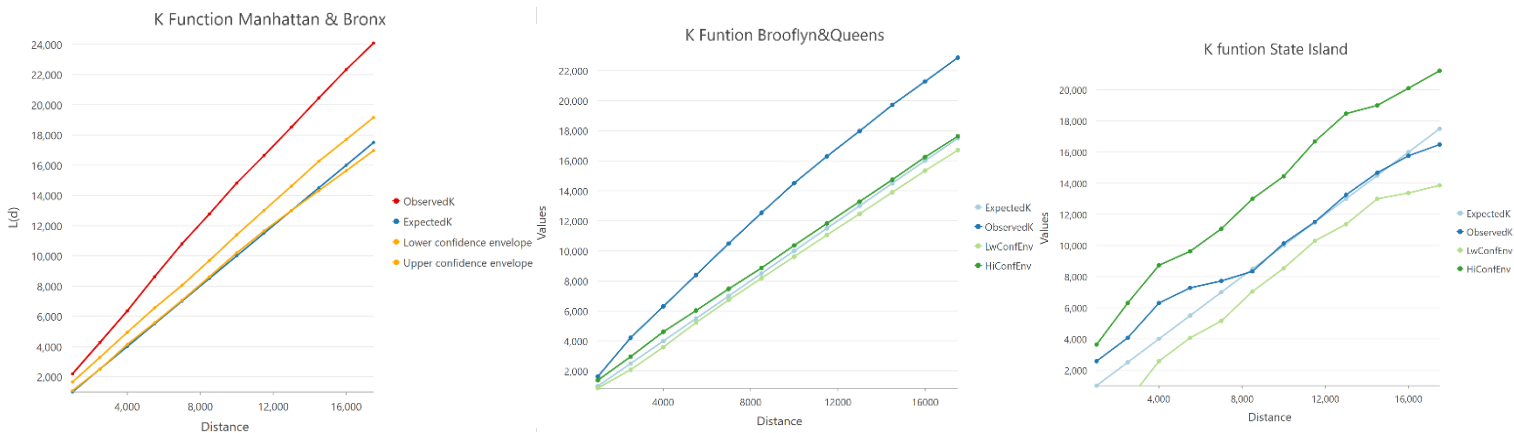


Figure 5 K Three groups

Inhomogeneity Adjustment

Inhomogeneity Adjustment

A key assumption of standard point pattern analysis is spatial homogeneity, meaning that the underlying population at risk is evenly distributed across the study area. In New York City, this assumption is clearly unrealistic. Population density varies dramatically across census tracts, with very high densities in parts of Manhattan and lower densities in Staten Island and peripheral areas. If raw incident counts are analyzed without accounting for this variation, areas with more people will mechanically appear to have more incidents, even if the underlying risk is not higher. For this reason, an inhomogeneity adjustment was necessary to separate population concentration from true spatial concentration of police violence incidents. Also, tracts with extremely small populations were handled carefully to avoid unstable rates.

To address this, incident counts were first aggregated to census tracts using a spatial join. Population-normalized rates were then calculated as incidents per 10,000 residents (FE_per10k). This rate surface represents risk relative to the population at risk rather than raw event density. However, rates alone do not indicate whether an area has more incidents than expected given its population share. Therefore, an expected value was calculated for each tract based on its proportion of the total NYC population:

$$\text{FE_per10k} = (\text{Join_Count} / \text{POP2020}) \times 10,000$$

Inhomogeneity Adjustment Incidents per 10k

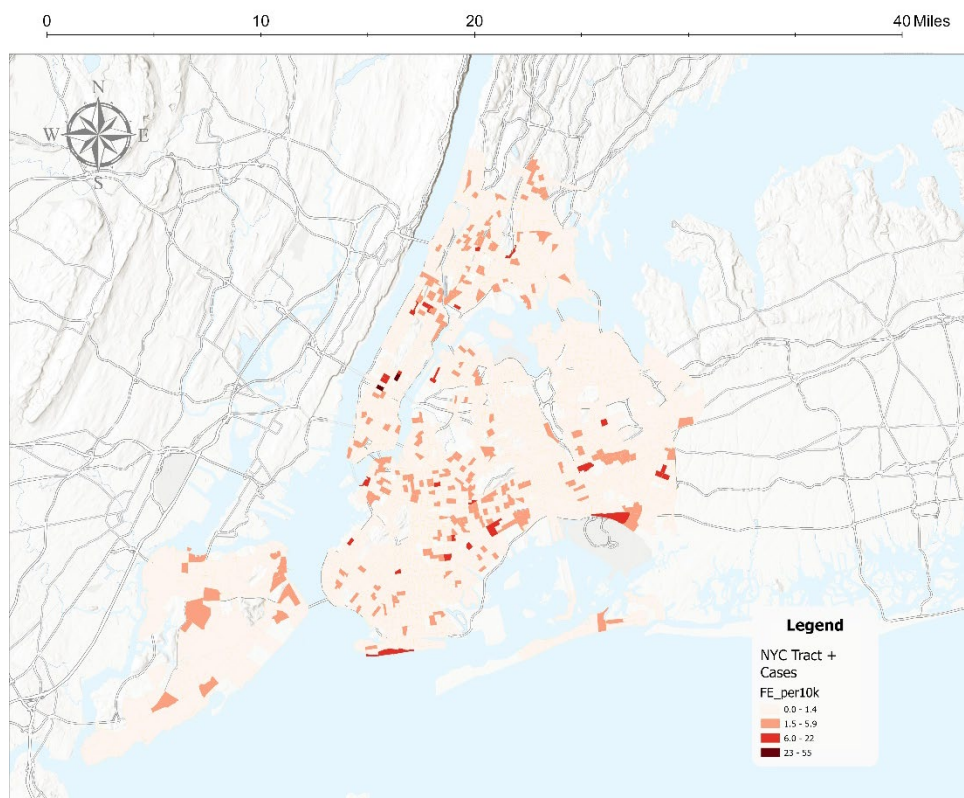


Figure 6 Map Incidents per10k

$\text{Expected_FE} = (\text{Tract Population} / \text{Total Population}) \times \text{Total Incidents}$

An excess measure was then derived: $\text{Excess_FE} = \text{Observed Incidents} - \text{Expected Incidents}$

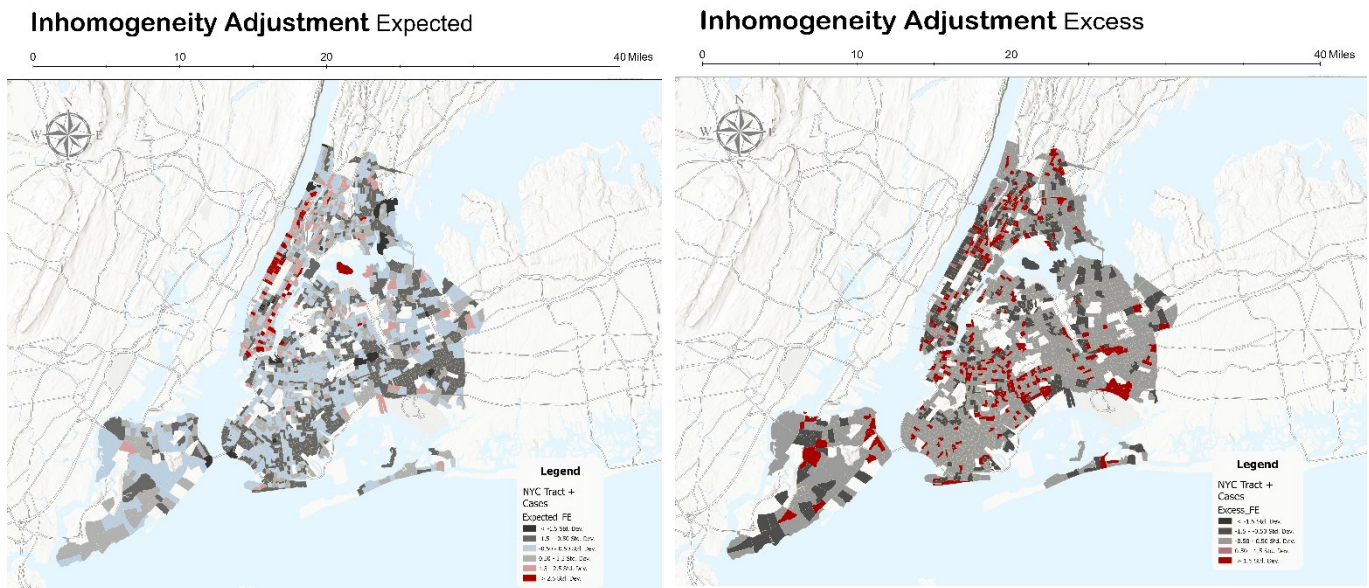


Figure 7 Map Expected vs Excess

. This framework allows comparison between what would be expected under proportional distribution and what was actually observed

The Expected map shows where incidents would occur if they were distributed strictly according to population share. As anticipated, high expected values concentrate in densely populated areas such as Manhattan and portions of Brooklyn and Queens. This map primarily reflects demographic structure rather than policing outcomes.

The Incidents per 10k map adjusts for tract population and reveals where incident rates are elevated relative to local population size. While many high-rate tracts overlap with high-density neighborhoods, several areas outside the densest cores also show elevated rates. This indicates that some locations experience disproportionately high incident rates even after adjusting for population.

The Excess map provides the clearest interpretation of spatial disproportionality. Red tracts represent areas where observed incidents exceed what would be expected given population share, while darker negative values represent areas with fewer incidents than expected. Concentrations of positive excess appear in parts of Upper Manhattan, segments of the Bronx, and selected areas in Brooklyn and Staten Island. These patterns suggest that clustering observed in the Ripley's K analysis is not solely a reflection of population density, but persists after controlling for demographic concentration.

Together, these maps demonstrate that spatial clustering of incidents in NYC cannot be fully explained by population distribution alone. The adjustment strengthens the validity of the spatial analysis by showing that certain neighborhoods exhibit disproportionately high incident counts relative to their expected baseline.

MAUP Sensitivity Analysis (Fishnet Approach)

A key limitation in spatial analysis of aggregated data is the Modifiable Areal Unit Problem (MAUP), which refers to the fact that analytical results can change depending on how spatial units are defined. Because census tracts vary substantially in size and shape across New York City, patterns observed in tract-level rates may partly reflect administrative boundaries rather than underlying spatial processes. To evaluate the robustness of our findings, a sensitivity test was conducted using a regular fishnet grid.

A uniform grid was generated over the NYC boundary using two resolutions: 2,000-foot and 4,000-foot cell sizes. The fishnet cells were clipped to the city boundary, and police encounter points were spatially joined to each grid cell to compute counts per cell. Population was allocated to grid cells using an area-weighted approach: census tracts were intersected with the fishnet, and each tract's population was proportionally redistributed based on the area of overlap. These redistributed population values were summed to the grid cell level, allowing calculation of incident rates per 10,000 residents for each cell.

The 2,000-foot fishnet map shows localized clusters of high encounter rates, particularly in portions of Upper Manhattan, the South Bronx, central Brooklyn, and parts of Staten Island. When the grid size increases to 4,000 feet, the pattern becomes smoother and more generalized, but

The primary high-rate areas persist. This persistence across grid resolutions indicates that the observed clustering is not solely an artifact of census tract boundaries. While the exact intensity and spatial extent of high-rate cells vary with grid size, as expected under MAUP, the core hotspots remain spatially stable.

MAUP (fishnet)

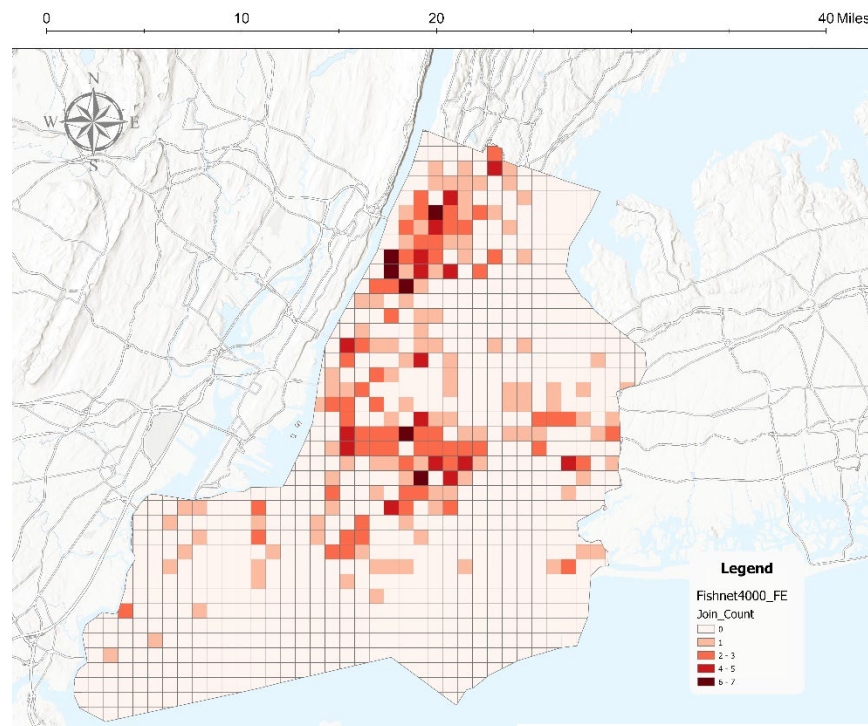


Figure 8 Fishnet 4000 + point count

MAUP (fishnet)

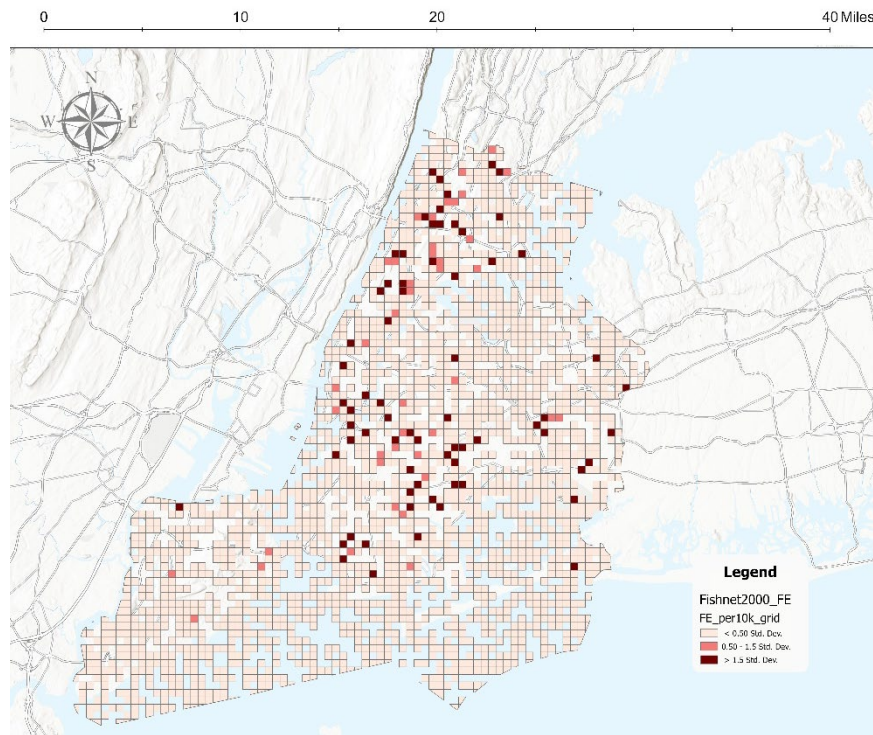


Figure 9 Fishnet 2000 (Cases per 10000 people)

This sensitivity analysis strengthens confidence in the study's conclusions. The similarity between tract-based inhomogeneity-adjusted maps and fishnet-based rate surfaces suggests that the identification of high-risk areas is not highly sensitive to zoning configuration. However, the observed differences in intensity between the 2,000-foot and 4,000-foot grids demonstrate that scale selection influences the apparent strength and concentration of clusters. Accordingly, results are interpreted as robust at the neighborhood-to-community scale rather than at very fine micro-scales.

Overall, the MAUP test confirms that spatial concentration patterns of police encounter rates in NYC are not an artifact of administrative tract boundaries and remain evident under alternative spatial aggregation schemes.

Results

The analyses provide consistent evidence that fatal police encounters in New York City are spatially clustered rather than randomly distributed. The global Ripley's K function shows the observed K curve exceeding the expected curve across all tested distance bands, and in multiple intervals surpassing the upper confidence envelope. This indicates statistically significant clustering across both neighborhood-scale and broader citywide distances.

When Ripley's K is applied to contiguous borough groupings, clustering remains evident but varies in magnitude. Manhattan and the Bronx demonstrate the strongest and most persistent deviations above the confidence envelope. Brooklyn and Queens also display statistically

significant clustering, though the intensity is comparatively more diffuse. Staten Island shows greater variability due to its smaller number of events, but the pattern does not approximate complete spatial randomness. These subgroup results confirm that clustering is not driven by a single borough but is distributed unevenly across the city.

Kernel Density Estimation at 3,000 ft, 6,000 ft, and 10,000 ft bandwidths reinforces the second-order findings. Smaller bandwidths reveal localized hotspots in Upper Manhattan, the Bronx, central Brooklyn, and select areas of Staten Island. As bandwidth increases, these hotspots consolidate into broader concentration zones, but the core areas persist. The stability of primary hotspot locations across bandwidths indicates that the intensity pattern is not dependent on a single smoothing parameter.

To address spatial inhomogeneity, incidents were normalized to the population at the census-tract level (incidents per 10,000 residents). Population adjustment modifies the apparent distribution of risk. Several high-population Manhattan tracts that appear prominent in raw clustering analyses show moderate per-capita rates once normalized. On the contrary, selected tracts in outer boroughs exhibit relatively elevated rates after adjustment.

The expected-versus-excess analysis further refines interpretation. Expected values estimate how many incidents would occur in each tract if cases were distributed strictly proportional to population. The excess map identifies tracts where observed counts exceed this expectation. Persistent excess clusters are visible in portions of Upper Manhattan, the Bronx, and central Brooklyn, indicating concentrations that are not fully explained by population distribution alone.

Finally, MAUP sensitivity testing using a 2,000-foot fishnet grid demonstrates that major concentration areas remain identifiable under an alternative zoning system. While intensity levels and boundaries shift modestly relative to tract-based aggregation, core hotspot regions remain evident. This suggests that the primary spatial patterns are reasonably robust to changes in areal unit configuration.

Collectively, global Ripley's K results, borough subgroup analyses, KDE surfaces, inhomogeneity adjustments, excess calculations, and MAUP testing converge in identifying structured spatial clustering across NYC.

Discussion

This study contributes to the growing body of spatial analysis literature demonstrating that fatal police encounters are not randomly distributed within urban environments. By integrating first-order intensity modeling (KDE) with second-order global clustering (Ripley's K) and population-adjusted rate analysis, the results extend beyond simple hotspot visualization and evaluate whether clustering persists after accounting for demographic structure.

The inhomogeneity adjustment is particularly important in this context. NYC is characterized by strong spatial variation in population density, land use, and socioeconomic conditions. Standard Ripley's K assumes spatial homogeneity under complete spatial randomness. Without adjustment, high event concentration in densely populated tracts could reflect exposure rather than interaction between events. By modeling population-normalized rates and comparing

observed to expected counts, this analysis demonstrates that certain areas exhibit elevated encounter levels even after demographic normalization. This strengthens the inference that clustering cannot be attributed solely to background population density (O’Sullivan & Unwin, 2010).

The MAUP sensitivity test further contributes methodological robustness. Spatial clustering studies are sensitive to the choice of aggregation unit (Openshaw, 1984). By recalculating rates using a regular fishnet grid, this study tests whether hotspot identification is dependent on census tract boundaries. Although magnitude and boundaries shift modestly, major concentration areas persist across zoning schemes. This cross-scale stability enhances confidence in the spatial findings and addresses a common limitation in urban spatial analysis.

Several limitations remain. First, the Fatal Encounters dataset is crowd-sourced and may contain underreporting, geocoding inaccuracies, or classification inconsistencies. Second, Ripley’s K relies on Euclidean distance, which may not fully capture network-based accessibility or physical barriers such as waterways. Third, census population is used as a proxy for exposure but does not incorporate policing intensity, socioeconomic disadvantage, or built environment variables that may influence spatial concentration. Future research could incorporate network-constrained distance metrics, temporally stratified analyses, or formal inhomogeneous K-function modeling to refine inference (Baddeley et al., 2000).

Policy implications should be approached cautiously. While excess clusters highlight areas warranting further investigation, spatial concentration does not imply causal explanation. However, identifying statistically significant concentration areas may support targeted inquiry into structural conditions, policing practices, and community-level characteristics (Braga et al., 2014; Weisburd, 2015). Spatial evidence thus functions as a diagnostic tool rather than a definitive explanatory model.

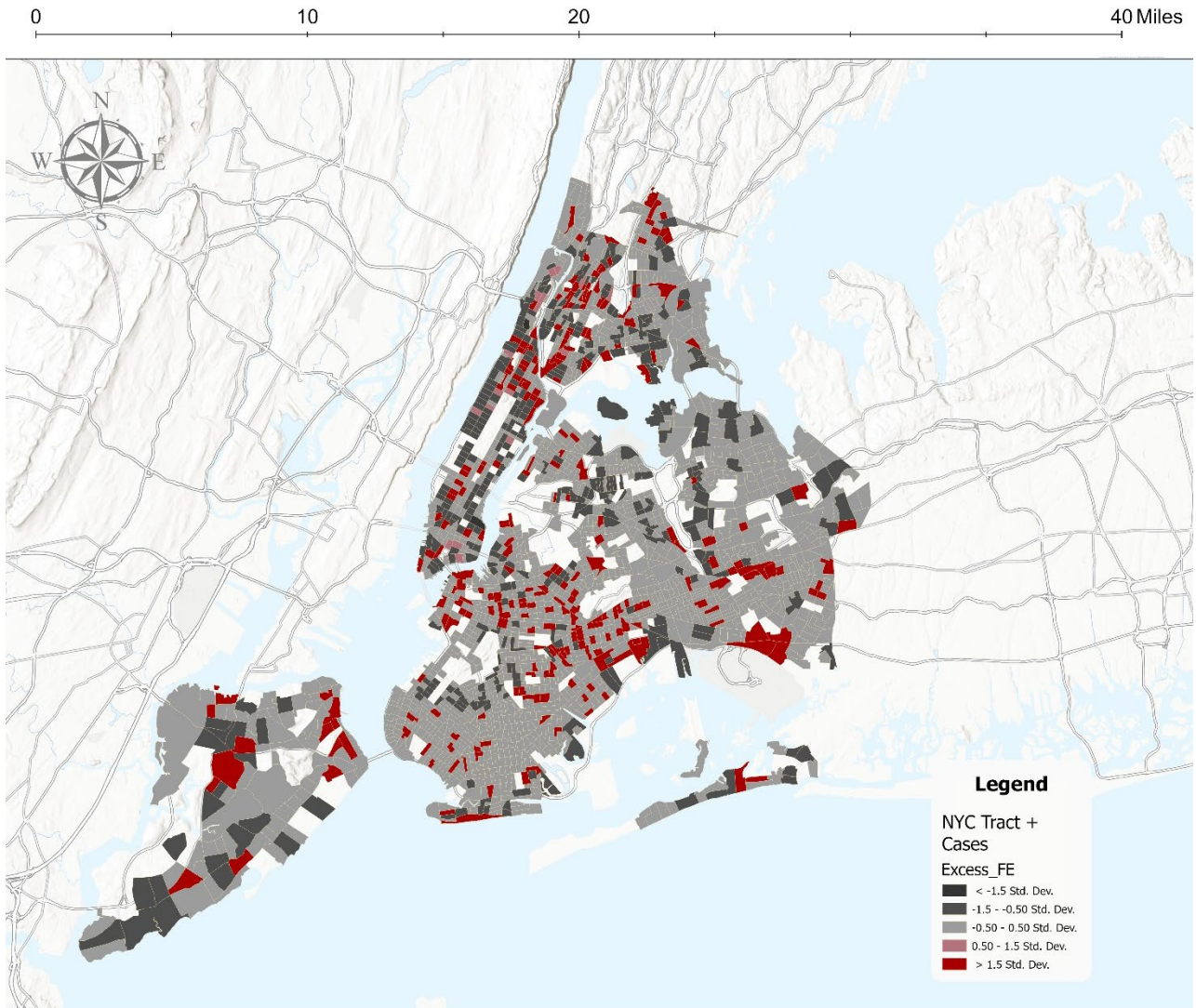
Overall, the integration of clustering diagnostics, demographic adjustment, and MAUP sensitivity testing provides a comprehensive framework for evaluating spatial concentration in heterogeneous urban environments while explicitly acknowledging methodological constraints.

References

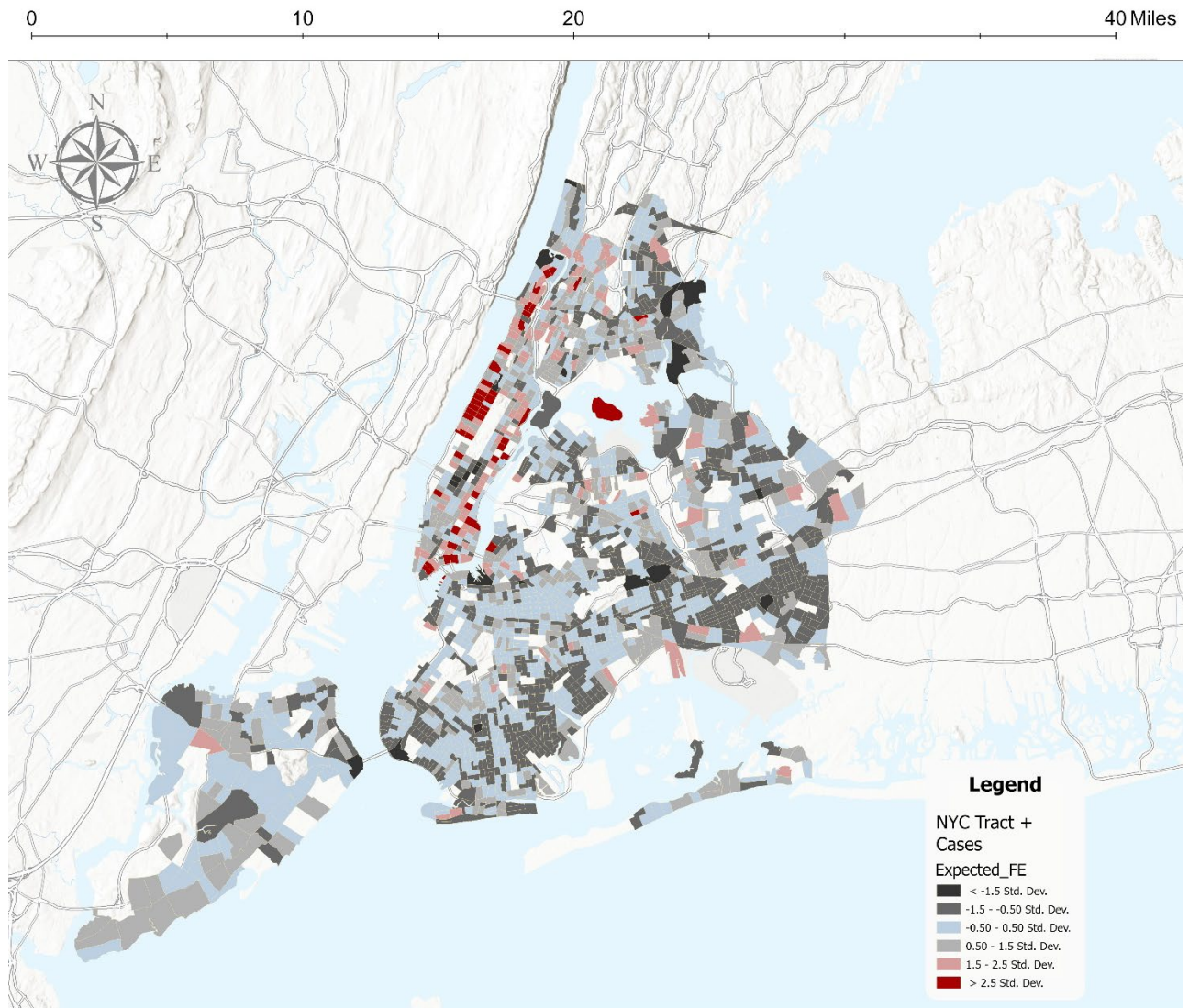
- Finch, B.K., Beck, A., Burghart, D.B., Johnson, R., Klinger, D., & Thomas, K. (2019). Using crowd-sourced data to explore police-related deaths in the United States (2000–2017): The Case of Fatal Encounters. *Open Health Data*, 6(1).
- O’Sullivan, D., & Unwin, D. (2010). *Geographic Information Analysis (2nd ed.)*. Wiley.
- Esri Documentation: Kernel Density; Multi-Distance Spatial Cluster Analysis; Create Fishnet.

Appendix.

Inhomogeneity Adjustment Excess



Inhomogeneity Adjustment Expected



MAUP (fishnet)

